

Phillip W. Yoon

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EDUCATION

Hunter College, City University of New York <i>M.A. in Mathematics</i>	2025–2027 (expected) New York, NY
University of Pennsylvania, Graduate School of Education <i>M.S.Ed. in Independent School Teaching Residency</i>	2022–2024 Philadelphia, PA
Northern Illinois University <i>Graduate (M.A.) coursework in philosophy, 18 credits (non-degree)</i>	2021–2022 DeKalb, IL
Johns Hopkins University <i>B.A. in Mathematics and Philosophy, General Honors</i> <i>Study abroad: Budapest Semesters in Mathematics, Summer 2019.</i>	2017–2021 Baltimore, MD

RESEARCH INTERESTS

Commutative and homological algebra, especially monomial ideals, Betti numbers, and combinatorial or topological methods in algebra. Secondary interests include noncommutative algebra, algebraic combinatorics, representation theory, and solvable lattice models.

PUBLICATIONS

1. Noah Ripke and **Phillip Yoon**, “Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables,” *Journal of Pure and Applied Algebra* **230** (2026), 108354.

RESEARCH EXPERIENCE

NSF REU/RET in Mathematics, California State University, Chico <i>Researcher, Commutative Algebra and Combinatorics Group</i>	June–August 2026 Chico, CA
• Conducted research under the supervision of Guillermo Alesandroni on characteristic dependence of Betti numbers of monomial ideals. • Proved with a coauthor that the multigraded, graded, and total Betti numbers of monomial ideals in five variables are independent of the characteristic of the base field; led the drafting and revision of the resulting manuscript. • Developed a homological and combinatorial proof based on Hochster’s formula; wrote Python code reducing 7,580 squarefree ideals to 210 representatives up to variable relabeling and carried out computational verification in Macaulay2.	
Hunter College, City University of New York <i>Independent Research in Combinatorics on Words and Noncommutative Algebra</i>	October 2025–present New York, NY
• Supervised by Be’eri Greenfeld; studying growth of associative and monomial algebras, Gelfand–Kirillov dimension, nilpotent ideals, and questions arising from combinatorics on words. • Presented this work at Hunter College’s student research colloquium.	
Polymath Jr., University of Minnesota <i>Research Participant</i>	June–August 2025 Remote
• Investigated recursive partition functions for lattice models related to quantum superalgebras under the mentorship of Ben Brubaker. • Contributed to a collaborative project involving color and supercolor variants and local recurrence relations; presented this work at the 2026 Joint Mathematics Meetings.	

RESEARCH TALKS AND PRESENTATIONS

“Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables” Invited talk , Route 81 Conference on Commutative Algebra and Algebraic Geometry, Syracuse University, Syracuse, NY.	September 2026
“What Does Small Growth Tell Us About Algebras?” Hunter Mathematics and Statistics Colloquium: Student Research Presentations, Hunter College, New York, NY.	April 2026
“Lattice Models for Quantum Superalgebras: Color and Supercolor” Joint Mathematics Meetings, Washington, DC	January 2026

Poster presentation, CUNY Directed Reading Program, The Graduate Center, CUNY, New York, NY.

ADDITIONAL MATHEMATICAL EXPERIENCE

CUNY Directed Reading Program, The Graduate Center

January–May 2025

Algebraic Geometry and Computational Commutative Algebra

New York, NY

- Read selected material from Cox, Little, and O’Shea’s *Ideals, Varieties, and Algorithms* with graduate mentor Valeriy Sergeev; completed an expository project and final presentation.

TEACHING EXPERIENCE

Hunter College, City University of New York

Fall 2026

Adjunct Lecturer, appointed

New York, NY

- Instructor of record for Precalculus (MATH 12550).

Columbia Grammar and Preparatory School

August 2025–June 2026

Teacher of Mathematics

New York, NY

- Courses taught: AP Calculus AB and Precalculus.

New York City College of Technology, CUNY

August–December 2024

Adjunct Lecturer

Brooklyn, NY

- Courses taught: College Algebra and Trigonometry (MAT 1275CO) and Precalculus (MAT 1375).

Trinity School

August 2022–June 2024

Mathematics Teaching Fellow

New York, NY

- Courses taught: Prealgebra and Honors Algebra I.

Johns Hopkins University

August 2018–May 2021

Teaching Assistant and Tutor

Baltimore, MD

- Supported Discrete Mathematics (EN.553.171), Probability and Statistics (EN.553.211), Calculus II (AS.110.109), and Linear Algebra (AS.110.201).

AWARDS AND HONORS

- John Arthur Loustau Scholarship, Hunter College, CUNY, 2026.
- Gail J. McGovern Scholarship, Johns Hopkins University, 2018–2020.
- Honorable Mention, International Olympiad on Astronomy and Astrophysics, 2016.

SERVICE AND OUTREACH

- Member, AI Committee, Columbia Grammar and Preparatory School, 2025–2026.
- MATHCOUNTS Coach, Trinity School, 2022–2024.
- Co-organizer, Johns Hopkins Mathematics Tournament, 2018–2020.

LANGUAGES AND TECHNICAL SKILLS

Languages: English, Korean, Japanese.

Technical: $\LaTeX$, Python, Macaulay2.

ADDITIONAL PROFESSIONAL EXPERIENCE

xAI

January–August 2025

AI Tutor

Remote

- Evaluated and improved AI-generated mathematical reasoning and written solutions.